

SEBI Grade A [IT] – PYQs

(Memory Based)

Q. In a data-warehouse, what is the primary purpose of building a data mart as compared to an enterprise warehouse?

- A) To store transactional data
- B) To serve a specific business line with focused subject area
- C) To replace operational systems
- D) To hold metadata only

Q. What will the following query return?

```
SELECT department_id, COUNT(*) FROM employees  
GROUP BY department_id HAVING COUNT(*) > 5;
```

- A) All employees in each department
- B) **Departments with more than 5 employees**
- C) Average employees per department
- D) Departments with exactly 5 employees

Q. A malware is inserted into a system such that it propagates automatically across the network, exploiting vulnerabilities without user action. Which of the following is the best description?

- A) Trojan horse
- B) Worm
- C) Virus
- D) Spyware

Q. An employee receives an email purporting to be from the company's IT team asking them to click a link and reset their password. The link leads to a fake login page that records credentials.

Which attack technique is this?

- A) **Phishing**
- B) Ransomware
- C) Drive-by Download
- D) Denial of Service

Q. In a switched LAN, what happens when a switch receives a frame with a destination MAC address that is *not* in its MAC address table?

- A) Frame is dropped
- B) Frame is forwarded out all ports except the receiving port
- C) Frame is sent to the router
- D) Frame is sent back to sender

Q. Which constraint ensures that values in a column are unique?

- A) PRIMARY KEY
- B) **UNIQUE**
- C) FOREIGN KEY
- D) NOT NULL



Q. You have a subnet 192.168.10.0/28. How many valid hosts can this subnet have?

- A) 16
- B) 14
- C) 12
- D) 30



Q. Which protocol is used for secure web communication?

- A) HTTP
- B) FTP
- C) **HTTPS**
- D) SMTP



Q. Given a function in Python:

```
def f(lst=[]):  
    lst.append(1)  
    return lst  
print(f(), f())
```

What is printed?

- A) [1], [1]
- B) [1], [1, 1]
- C) [1, 1], [1, 1]
- D) [1], [1, 1, 1]

Q. In a directed, weighted graph with possible negative weights but no negative cycles, you want to compute shortest paths from a source. Which algorithm is appropriate?

- A) Dijkstra's
- B) Bellman-Ford
- C) Kruskal's
- D) Prim's

Q. In operating systems, “Thrashing” occurs when:

- A) Processes are waiting for I/O devices
- B) CPU utilization is high
- C) Page faults occur excessively due to low memory
- D) The system deadlocks

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Q. In the OSI model, the layer responsible for establishing, managing and terminating sessions between applications is:

- A) Transport layer
- B) Session layer
- C) Data Link layer
- D) Presentation layer

Q. Which of the following sorting algorithms has the best worst-case time complexity among the list?

- A) Bubble sort
- B) Insertion sort
- C) Merge sort
- D) Selection sort

Q. In a data cube, users often perform drill-down or roll-up operations. If you move from “year” to “quarter” to “month”, you are performing what?

- A) Roll-up
- B) Drill-up
- C) **Drill-down**
- D) Slice

Q. In the context of relational databases, the process of removing redundant data and ensuring data dependencies are properly enforced is called:

- A) Denormalization
- B) Normalization
- C) Indexing
- D) Partitioning

Q. Which of the following concurrency control protocols allows non-serializable but recoverable schedules?

- A) Two-phase locking (2PL)
- B) Timestamp ordering
- C) **Optimistic concurrency control**
- D) Deadlock prevention

**Q. A ransomware attack encrypts all data on a corporate server and displays a message demanding Bitcoin payment to decrypt.
Which of the CIA triad (Confidentiality, Integrity, Availability) is most directly impacted?**

- A) Confidentiality only
- B) Integrity only
- C) Availability only
- D) All three

Q. In Python, the slicing expression `my_list[2:8:2]` means:

- A) **Select every second element from index 2 to 8 (exclusive)**
- B) Select all elements between 2 and 8 inclusive
- C) Select only elements at index 2 and 8
- D) Reverse the list



Q. To open a file for both reading and writing without truncating it, which mode should be used?

- A) 'r'
- B) 'w'
- C) 'a'
- D) 'r+' ☒

Q. What does the following regex pattern match?

```
r'^[A-Z][a-z]+$'
```

- A) A string containing all uppercase letters
- B) A string starting with an uppercase letter followed by lowercase letters
- C) Any lowercase string
- D) Only a single uppercase character

Q. What is the correct syntax to display the value of a variable in shell?

- A) `echo $var`
- B) `print(var)`
- C) `display var`
- D) `echo var`

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Q. A gateway operates at which OSI layer(s)?

- A) Network layer
- B) Transport layer
- C) Multiple layers
- D) Data Link layer

Q. Which UNIX command is used to count the number of lines, words, and characters in a file?

- A) count
- B) cat
- C) **wc**
- D) grep

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Q. FTP uses two ports: one for control and one for data. Which of the following is correct?

- A) 20 for control, 21 for data
- B) 21 for control, 20 for data
- C) Both use 25
- D) Both use 80

Q. To search for the word “error” in a log file (case-insensitive), which command is correct?

- A) `grep “error” logfile`
- B) `grep -i “error” logfile`
- C) `find “error” logfile`
- D) `awk “error” logfile`

Q. Which of the following steps ensures that inconsistent, duplicate, and missing data are handled properly?

- A) Data Extraction
- B) **Data Cleaning**
- C) Data Loading
- D) Data Transformation

Q. The process of extracting data from multiple heterogeneous sources and storing it in a data warehouse is known as:

- A) Data Integration
- B) Data Mining
- C) **ETL**
- D) OLAP

Q. In a Data Cube, the term *dimension* refers to:

- A) Measurable numeric data
- B) Attribute or perspective for analysis
- C) Metadata storage
- D) Schema definition

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Q. In a shell script, \$0 refers to:

- A) The first argument
- B) **The name of the script itself**
- C) The last argument
- D) Total number of arguments

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Q. The `chmod 755 file.sh` command does what?

- A) Gives full permission to everyone
- B) Gives read+write+execute to owner, read+execute to others
- C) Read-only to owner
- D) Execute only



```
void fun() {  
    static int x = 0;  
    x++;  
    printf("%d ", x);  
}  
  
int main() {  
    fun();  
    fun();  
    fun();  
}
```

Q. In C, What will be the output of the above program?

- A) 1 1 1
- B) 1 2 3
- C) 3 3 3
- D) 0 1 2

Q. Which firewall type inspects both the header and the payload of packets?

- A) Packet-filtering firewall
- B) Stateful inspection firewall
- C) Application (Proxy) firewall
- D) Circuit-level gateway

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Q. In DFS of a graph, when a vertex has no unvisited adjacent vertices, we —

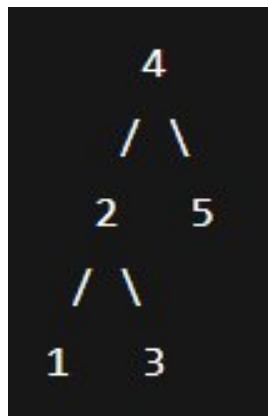
- A) Continue exploring
- B) **Backtrack**
- C) Restart DFS
- D) Stop immediately

```
using namespace std;  
int x=50;  
int  
    main(){ int  
        x=10;  
        cout<<::x;  
    }
```

Q. In C++, What will be the output of the above program?

- A) 10
- B) 50
- C) Compile Error
- D) 0

Q. Inorder traversal of the following binary tree produces what sequence?



- A) 4 2 1 3 5
- B) 1 2 3 4 5
- C) 1 5 3 2 1
- D) 2 4 3 2 1

Q. Number of spanning trees in a complete graph with 5 vertices (K_5)?

- A) 10
- B) 25
- C) 125
- D) 64



Q. Level order traversal of the tree below is:



- A) **A B C D E F**
- B) D B E A F C
- C) A C B D F E
- D) D E F B C A

Q. If a binary tree has N nodes, how many edges does it have?

- A) N
- B) $N-1$
- C) $N+1$
- D) $2N$



Q. The number of edges traversed in DFS of a connected graph with V vertices and E edges is —

- A) V
- B) **E**
- C) $V + E$
- D) Depends on graph type

Q. Which of the following statements is true for Minimum Spanning Tree (MST)?

- A) MST is unique for all graphs
- B) MST may not be unique if equal edge weights exist
- C) MST always includes all edges
- D) MST always has maximum weight

```
int fact(int n){  
    if(n<=1) return 1;  
    else return n*fact(n-1);  
}  
int main(){  
    printf("%d", fact(4));  
}
```

Q. What will be the output of the above program?

- A) 12
- B) 24
- C) 16
- D) 20

Q. Which of the following is TRUE about firewalls?

- A) Firewalls prevent all types of attacks
- B) Firewalls can block unauthorized traffic based on rules
- C) Firewalls encrypt all transmitted data
- D) Firewalls replace antivirus software